

Pooja Kiran

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EDUCATION

Arizona State University

Master of Science in Information Technology (Security) – GPA: 3.87/4.00

Tempe, AZ

Aug. 2024 – May 2026

M. S. Ramaiah University of Applied Sciences

Bachelor of Technology in Computer Science and Engineering – CGPA: 8.44/10

Bengaluru, India

Aug. 2019 – Aug. 2023

EXPERIENCE

Independent AI Security Researcher & Engineer

Aug. 2024 – Present

Self-Directed Research

Tempe, AZ

- Built and evaluated security controls across agent runtime enforcement, AWS IAM, LLM red teaming, model supply-chain analysis, training-data integrity, and adversarial ML, with threat models and reproducible evidence across 19 repositories.
- Hardened repositories with protected main, CI quality gates, SARIF/Code Scanning workflows, dependency/security checks, and full-history secret scanning; final Gitleaks verification returned zero findings across all 19 repositories.
- Integrated FastAPI, Docker, Elastic Security telemetry, multi-version Python CI, and evidence-backed runbooks/posters so security behavior, failure modes, and limitations remain reviewable and repeatable.

Business & Compliance Lead – AEROSSEC

Aug. 2025 – Dec. 2025

Honeywell Aerospace Technologies × Arizona State University

Tempe, AZ

- Led business/compliance analysis for a proposed airline PSS protection layer and built a \$120K first-year commercialization scenario with a five-year financial model presented to ASU and Honeywell stakeholders.

Graduate Teaching Assistant – IT Grader

Jan. 2025 – Oct. 2025

Ira A. Fulton Schools of Engineering, Arizona State University

Mesa, AZ

- Evaluated approximately 85 undergraduate web-development submissions per semester and 52 graduate cybersecurity assignments covering policy, compliance, risk analysis, and security controls.

SELECTED SECURITY PROJECTS

MCP Agent Security Gateway | Python, FastAPI, MCP/JSON-RPC, Elastic Security

Jul. 2026 – Present

- Built an inline MCP/JSON-RPC security gateway with 55 prompt-injection patterns, capability checks, PII/exfiltration signals, rate limiting, default-deny policy enforcement, anti-SSRF controls, and hash-chained audit evidence.
- Validated 718 automated tests at 82.46% statement coverage, with the same suite green on Python 3.10–3.12 and fail-closed tests proving denied calls do not reach downstream transport.
- Implemented 9 Elastic Security rules and 21 core SIEM tests; added ATT&CK/ATLAS mappings, synthetic lifecycle evaluation, and security telemetry for investigation and tuning.

AWS Agent Identity Guard | Python, AWS IAM, SARIF 2.1.0

Aug. 2026 – Sep. 2026

- Built a static IAM analyzer with 25 deterministic rules covering wildcard access, `iam:PassRole`, `sts:AssumeRole`, privilege escalation, risky trust policies, audit tampering, and missing permission boundaries.
- Emitted text, JSON, and SARIF 2.1.0 for GitHub Code Scanning, with CI exit enforcement on critical/high findings and zero runtime dependencies for local-file scans.
- Verified 238 collected tests with 235 passing and 3 credential-gated live-scan skips; positive/negative tests cover every emitted rule ID plus parser fuzzing and failure modes.

HF Model Provenance Scanner | Python, SafeTensors, GGUF, ONNX, Keras, SARIF

Jul. 2026 – Sep. 2026

- Built a non-executing model supply-chain scanner combining pickle-opcode parsing, AST/taint/symbolic-string analysis, provenance checks, dependency risk, and binary-format inspection without loading untrusted model code.
- Validated 241 passing tests, 1 platform-specific skip, 6 additional pytest subtests, and 75.81% statement coverage across Python 3.10–3.12 CI.
- Detected 33/33 committed adversarial fixtures with 0 actionable findings across 4 committed benign samples; documented these as fixture-scoped regression results, not universal detection claims.

Dataset Poisoning Detector | Python, FastAPI, Redis Streams, Kafka, Prometheus

Jul. 2026 – Present

- Built a training-data security pipeline combining statistical outlier checks, Isolation Forest, spectral and label-aware signals, quarantine, dead-letter routing, and streaming backpressure controls.
- Hardened API and streaming boundaries with fail-closed authentication, request-size limits, shared production rate limiting, trusted-baseline readiness, bounded scoring, and WebSocket abuse controls.
- Verified 199 passing tests at 90.88% statement coverage with a 90% CI/release gate, including Redis/Kafka runtime paths, readiness failures, rate limiting, and error handling.

TECHNICAL SKILLS

Languages: Python, Rust, C++ | **Security:** AI/Agent Security, MCP/JSON-RPC, Application Security, AWS IAM, LLM Red Teaming, Model Supply-Chain Security, Detection Engineering

Tooling: FastAPI, pytest, Hypothesis, GitHub Actions, SARIF, CodeQL, Bandit, Trivy, Docker, Elastic Security, Prometheus | **Certifications:** AWS Academy Cloud Architecting; AWS Academy Cloud Security Foundations